

A Short Guide to Lemma C.4

Translation-covariant scalar kernels are convolutions

Abstract

Lemma C.4 identifies every positive translation-covariant map from $L^1(\mathbf{E})$ to finite measures on a finite-dimensional Euclidean space $\mathbf{E}$. Such a map must first dilate the input location by $\sqrt{\gamma}$ and then add noise with a fixed finite measure μ . The proof obtains μ by feeding the map an approximate point mass at the origin. This note explains the covariance convention, the weak-* extraction, and the passage from approximate point masses to arbitrary L^1 inputs.

1 The statement in probabilistic language

The lemma is stated on a finite-dimensional Euclidean space $\mathbf{E}$, including $\mathbf{E} = \{0\}$. The zero-dimensional case is immediate, so below assume $\dim \mathbf{E} > 0$ and equip $\mathbf{E}$ with Lebesgue measure dx . We use $E \subseteq \mathbf{E}$ for a Borel set. If $u \geq 0$ and $\int u(x) dx = 1$, then $u(x) dx$ is the law of an input random variable X . Recall that $\mathcal{R}$ is the positive contraction from Lemma C.4,

$$\mathcal{R} : L^1(\mathbf{E}) \longrightarrow C_0(\mathbf{E})^*.$$

By the Riesz representation theorem, the positive functional $\mathcal{R}(u)$ is represented by the unique positive Radon measure ν_u satisfying

$$\langle f, \nu_u \rangle := \mathcal{R}(u)(f) = \int_{\mathbf{E}} f(y) \nu_u(dy) \quad (f \in C_0(\mathbf{E})).$$

Contractivity gives $\nu_u(\mathbf{E}) = \|\nu_u\| \leq \|u\|_1 = 1$; hence it is finite. If $\nu_u(\mathbf{E}) = 1$, one may choose an output random variable Y with law ν_u , so that $\Pr(Y \in E) = \nu_u(E)$. If $c_u := \nu_u(\mathbf{E}) < 1$, then ν_u is instead a subprobability measure; it can be represented by adjoining a failure point $\partial \notin \mathbf{E}$ and setting

$$\Pr(Y \in E) = \nu_u(E), \quad \Pr(Y = \partial) = 1 - c_u.$$

To match the manuscript's notation, write $T_a \nu$ for the translate of a finite measure ν by a :

$$(T_a \nu)(E) := \nu(E - a), \quad E - a := \{y - a : y \in E\}.$$

If ν is the law of Y , then $T_a \nu$ is the law of $Y + a$. Likewise, $u(\cdot - h)$ is the density of $X + h$. Hence the covariance assumption of Lemma C.4 is the equality of finite Radon measures

$$\mathcal{R}(u(\cdot - h)) = T_{\sqrt{\gamma}h} \mathcal{R}(u) \quad (u \in L^1(\mathbf{E}), h \in \mathbf{E}). \tag{1}$$

Evaluating this identity on a Borel set E gives the more concrete formula

$$\mathcal{R}(u(\cdot - h))(E) = \mathcal{R}(u)(E - \sqrt{\gamma}h).$$

Thus replacing the input law of X by that of $X + h$ replaces the output measure ν_u by its translate through $\sqrt{\gamma}h$; when the measures are probability laws, this says that the new output has the same law as $Y + \sqrt{\gamma}h$.

Lemma C.4 says that there is one finite positive measure μ such that

$$\mathcal{R}(u)(E) = \int_{\mathbf{E}} u(x) \mu(E - \sqrt{\gamma}x) dx. \quad (2)$$

For each input point x , define the finite measure

$$K(x, E) := \mu(E - \sqrt{\gamma}x).$$

Then (2) says that $\mathcal{R}(u)$ is the mixture of the pointwise output measures $K(x, \cdot)$ with weights $u(x) dx$. If $\mu(\mathbf{E}) = 1$, each $K(x, \cdot)$ is a probability measure and may be read as the conditional law of the output given $X = x$:

$$K(x, E) = \Pr(Y \in E \mid X = x).$$

Indeed, if $Z \sim \mu$ is independent of X and $Y := \sqrt{\gamma}X + Z$, then

$$\Pr(Y \in E \mid X = x) = \mu(E - \sqrt{\gamma}x) = K(x, E),$$

and averaging the conditional measures over X gives

$$\Pr(Y \in E) = \int_{\mathbf{E}} u(x) K(x, E) dx = \mathcal{R}(u)(E).$$

The reason covariance forces the noise measure to be independent of x is most transparent at the level of point inputs. Temporarily suppose that δ_x were an admissible input and denote its output measure by K_x ; this notation does not assume the desired representation. Covariance would give $K_{x+h}(E) = K_x(E - \sqrt{\gamma}h)$. Recenter K_x by defining

$$\mu_x := T_{-\sqrt{\gamma}x} K_x, \quad \mu_x(E) = K_x(E + \sqrt{\gamma}x).$$

If K_x is a probability law, then μ_x is precisely the law of the residual $Y - \sqrt{\gamma}x$. The covariance identity now gives

$$\mu_{x+h}(E) = K_{x+h}(E + \sqrt{\gamma}(x + h)) = K_x(E + \sqrt{\gamma}x) = \mu_x(E).$$

Since h is arbitrary, all the recentered measures μ_x coincide. Point masses do not belong to $L^1(\mathbf{E})$, so the proof makes this heuristic rigorous by approximating δ_0 , extracting its output limit μ , and extending the formula to all L^1 inputs by convolution.

Finally, if $c := \mu(\mathbf{E}) < 1$, then $K(x, \mathbf{E}) = c$ for every x and $\mathcal{R}(u)(\mathbf{E}) = c$ for every normalized u . When $0 < c < 1$, this may be realized by taking a Bernoulli variable S with $\Pr(S = 1) = c$, an independent noise variable Z with law μ/c , and declaring the output to be $\sqrt{\gamma}X + Z$ when $S = 1$ and ∂ when $S = 0$. In the application, the missing mass $1 - c$ records probability that may have escaped to infinity in the weak-* limit.

2 Functional-analytic setting

The output space is

$$C_0(\mathbf{E})^*,$$

which the Riesz representation theorem identifies with the finite Radon measures on $\mathbf{E}$. The dual pairing of a test function f with a measure ν is

$$\langle f, \nu \rangle := \int_{\mathbf{E}} f(y) \nu(dy).$$

When ν is the law of a random variable Y , this pairing is $\mathbb{E}[f(Y)]$; for a subprobability measure, the same interpretation holds after setting $f(\partial) = 0$ at the failure point. Weak-* convergence of measures means

$$\int f d\mu_j \longrightarrow \int f d\mu \quad (f \in C_0(\mathbf{E})).$$

Thus weak-* convergence asserts convergence of these test-function averages, not pointwise convergence of random variables.

The map

$$\mathcal{R} : L^1(\mathbf{E}) \longrightarrow C_0(\mathbf{E})^*$$

is a positive contraction. Hence, for $u \geq 0$,

$$\|\mathcal{R}(u)\| \leq \|u\|_1.$$

For a positive measure its norm equals its total mass, so a normalized nonnegative input produces an output measure of mass at most one.

3 Statement of Lemma C.4

Lemma C.4 (rephrased). *Let $\mathcal{R} : L^1(\mathbf{E}) \rightarrow C_0(\mathbf{E})^*$ be a positive contraction satisfying (1). Then a finite positive measure μ with $\mu(\mathbf{E}) \leq 1$ exists such that (2) holds for every $u \in L^1(\mathbf{E})$ and every Borel set E .*

4 Proof walkthrough

Step 1: probe the kernel at the origin

Choose a nonnegative approximate identity $u_j \in C_c(\mathbf{E})$:

$$\int_{\mathbf{E}} u_j(x) dx = 1, \quad \text{supp}(u_j) \longrightarrow \{0\}.$$

The measures $u_j(x) dx$ are probability laws concentrating at the input point 0, so they approximate the point mass δ_0 . Heuristically, the output produced by δ_0 is the noise measure; since δ_0 is not an L^1 density, use instead the sequence of output measures

$$\mu_j := \mathcal{R}(u_j).$$

Thus μ_j is a measure, not a random variable; when it has mass one, one may represent it as the law of an output random variable. Positivity and contractivity imply $\mu_j(\mathbf{E}) \leq 1$.

The unit ball of $C_0(\mathbf{E})^*$ is weak-* compact. Since $C_0(\mathbf{E})$ is separable, a subsequence converges weak-* to a positive measure μ :

$$\int f d\mu_j \longrightarrow \int f d\mu \quad (f \in C_0(\mathbf{E})). \quad (3)$$

The norm is weak-* lower semicontinuous, so $\mu(\mathbf{E}) \leq 1$. Equality need not hold because the constant function 1 does not belong to $C_0(\mathbf{E})$ when $\mathbf{E}$ is noncompact.

Step 2: smooth an arbitrary input

Use the convolution convention

$$(u * u_j)(y) = \int_{\mathbf{E}} u(x) u_j(y - x) \, dx = \int_{\mathbf{E}} u(x) u_j(\cdot - x)(y) \, dx.$$

For normalized nonnegative u , if X has density u and V_j has density u_j independently of X , then $u * u_j$ is the density of $X + V_j$. Equivalently, it mixes the translated densities $u_j(\cdot - x)$ with weights $u(x) \, dx$; the proof then extends the identity to arbitrary $u \in L^1(\mathbf{E})$ by linearity. The approximate-identity property gives

$$u * u_j \longrightarrow u \quad \text{in } L^1(\mathbf{E}). \quad (4)$$

Bounded linearity of $\mathcal{R}$ therefore implies

$$\mathcal{R}(u * u_j) \longrightarrow \mathcal{R}(u)$$

in the norm of finite measures.

Because $u * u_j = \int_{\mathbf{E}} u(x) u_j(\cdot - x) \, dx$, bounded linearity moves $\mathcal{R}$ through this integral, while covariance gives $\mathcal{R}(u_j(\cdot - x)) = T_{\sqrt{\gamma}x} \mathcal{R}(u_j)$. Hence, as an equality of finite Radon measures,

$$\mathcal{R}(u * u_j) = \int_{\mathbf{E}} u(x) \mathcal{R}(u_j(\cdot - x)) \, dx = \int_{\mathbf{E}} u(x) T_{\sqrt{\gamma}x} \mathcal{R}(u_j) \, dx.$$

Evaluating this identity on a Borel set E gives

$$\begin{aligned} \mathcal{R}(u * u_j)(E) &= \int_{\mathbf{E}} u(x) \mathcal{R}(u_j(\cdot - x))(E) \, dx \\ &= \int_{\mathbf{E}} u(x) \mathcal{R}(u_j)(E - \sqrt{\gamma}x) \, dx. \end{aligned} \quad (5)$$

Thus the desired convolution formula already holds for the smoothed input $u * u_j$, with the finite measure $\mu_j = \mathcal{R}(u_j)$ in place of μ .

Step 3: pass to the weak-* limit

To justify the limit in (5), pair both sides with an arbitrary $f \in C_0(\mathbf{E})$. The right side becomes

$$\int_{\mathbf{E}} u(x) \left(\int_{\mathbf{E}} f(r + \sqrt{\gamma}x) \mu_j(\,dr) \right) \, dx. \quad (6)$$

Here r is the integration variable for the measure μ_j ; if μ_j is a probability law and $Z_j \sim \mu_j$, the inner integral is $\mathbb{E}[f(\sqrt{\gamma}x + Z_j)]$. For each fixed x , the translated function $r \mapsto f(r + \sqrt{\gamma}x)$ still belongs to $C_0(\mathbf{E})$, so (3) gives convergence of the inner integral to the same expression with μ . Its absolute value is bounded by $\|f\|_{\infty}$, and $|u|$ is integrable. Dominated convergence therefore turns (6) into

$$\int_{\mathbf{E}} u(x) \left(\int_{\mathbf{E}} f(r + \sqrt{\gamma}x) \mu(\,dr) \right) \, dx.$$

On the other side, norm convergence of the output measures gives

$$\int f \, d\mathcal{R}(u * u_j) \longrightarrow \int f \, d\mathcal{R}(u).$$

The two limiting test-function integrals are therefore equal for every $f \in C_0(\mathbf{E})$. For each fixed x , the inner integral is the pairing of f with the translated measure $T_{\sqrt{\gamma}x}\mu$, because

$$\int_{\mathbf{E}} f(r + \sqrt{\gamma}x) \mu(dr) = \int_{\mathbf{E}} f(y) (T_{\sqrt{\gamma}x}\mu)(dy), \quad (T_{\sqrt{\gamma}x}\mu)(E) = \mu(E - \sqrt{\gamma}x).$$

Since functions in $C_0(\mathbf{E})$ separate finite Radon measures, the matching measure identity is therefore, for every Borel set E ,

$$\mathcal{R}(u)(E) = \int_{\mathbf{E}} u(x) \mu(E - \sqrt{\gamma}x) dx,$$

which is the measure identity (2).

5 Examples

1. If $\mu = \delta_0$, then

$$\mathcal{R}(u)(E) = \int u(x) \mathbf{1}_E(\sqrt{\gamma}x) dx,$$

so for $X \sim u(x) dx$ the output variable is $Y = \sqrt{\gamma}X$.

2. If μ is a probability measure and $Z \sim \mu$ is independent of X , then the output measure $\mathcal{R}(u)$ is the law of $Y = \sqrt{\gamma}X + Z$.
3. If $\mu(\mathbf{E}) = c < 1$, then $\mathcal{R}(u)$ is a subprobability measure of mass c for every normalized u ; in the failure-point representation, the output lies in $\mathbf{E}$ with probability c and equals ∂ with probability $1 - c$.

No density for μ is required. The noise may have atoms or a singular part, which is important because weak-* limits of absolutely continuous output measures need not remain absolutely continuous.

6 Role in Proposition C.5

In the full classical–quantum specialization, $\mathcal{R}$ is the limiting classical trace kernel of Følner-averaged channels. Lemma C.4 isolates all its freedom in the one measure μ . The classical Gaussian witness then yields

$$\int c_{\gamma}(r) \mu(dr) \leq B_{\text{cl}}(\gamma, d) \mu(\mathbf{E}) \leq B_{\text{cl}}(\gamma, d).$$

That bound controls the trace of the weighted quantum limit and supplies the classical factor in the final hybrid fidelity bound.